

Keeno Smith

Full-Stack Software Engineer & AI Engineer

AI Application Development | Generative & Agentic AI | Cloud Engineering | DevOps & Automation

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Professional Summary

Full-Stack Software Engineer and Web Developer specialised in the design, development, integration, and deployment of end-to-end software systems across frontend applications, backend services, REST APIs, databases, authentication, system integration, and application infrastructure. My experience combines formal software engineering and computer science education with extensive hands-on product development, independent engineering, and technical delivery across the full software lifecycle.

My engineering work increasingly centres on AI application development, including LLM integration Retrieval-Augmented Generation (RAG), semantic search, embedding pipelines, prompt and context engineering, agentic architectures, tool calling, and AI-driven workflows. I build the surrounding application and data infrastructure required to make AI systems useful in practice, including document processing, retrieval pipelines, structures outputs, API integrations, evaluation, state management, and controlled execution.

I work across AWS, Microsoft Azure, and Google Cloud, developing cloud-native and distributed systems using serverless and containerised workloads, event-driven architectures, CI/CD pipelines, infrastructure services, security controls, observability, and production-oriented deployment patterns. Alongside cloud engineering, I develop event-driven workflow and integration systems with n8n, combining APIs, webhooks, asynchronous processing, orchestration, reliability controls, monitoring, and custom application interfaces to connect and automate business and technical processes.

Technical Skills

Programming Languages

JavaScript · Python · SQL · TypeScript · HTML · CSS

Frontend & Backend Engineering

React · Vite · Node.js · Express.js · REST APIs · API Design · Authentication & Authorisation · OpenAPI

AI Application Development

Generative AI · LLM Integration · RAG · Embeddings · Vector Search · Prompt & Context Engineering · Agentic AI · Multi-Agent Systems · Tool Calling

Cloud & AI Platforms

Microsoft Azure · Microsoft Foundry · Azure OpenAI · AWS · Amazon Bedrock · Bedrock AgentCore · GCP · Vertex AI · Serverless · Containers · Event-Driven Architecture

DevOps, Automation & SRE

Docker · Git · GitHub · GitHub Actions · CI/CD · n8n · Workflow Orchestration · Infrastructure & Deployment Automation · Observability · Incident Response

Data & Security

MongoDB · PostgreSQL · Mongoose · Prisma · Database Design · IAM · RBAC · Least Privilege · Cloud Security · Security Automation

Education

June 2026 - September 2026

Stellenbosch University - *Generative AI*

Generative AI training focused on modern AI concepts, generative models, AI application development, and practical applications of emerging AI technologies.

January 2026 - June 2026

HyperionDev - *Full-Stack Software Engineering & Web Development*

Full-stack software engineering and web development training covering frontend, backend, databases, APIs, software architecture, and application deployment.

January 2017 - January 2022

University of South Africa - *BSc Computer Science*

Computer Science degree studies covering core programming, software development, computer science, and related technical disciplines

January 2017 - January 2022

Greenside High School - *National Senior Certificate*

Experience

January 2022 - Present

Software Engineering & Application Development

Independently delivered full-stack software systems from requirements analysis and solution design through architecture, implementation, integration, testing, deployment, and ongoing maintenance. Held direct ownership of technical decisions across the software lifecycle, with responsibility spanning frontend and backend engineering, API and database architecture, authentication, system integration, AI-enabled functionality, production delivery, and continuous improvement.

- **Full-Stack Software Engineer**

Authenticated MERN Workspace Application

Designed and developed an end-to-end productivity platform using React, Node.js / Express, and MongoDB, with JWT authentication, interconnected resources, account lifecycle management, and cloud deployment. The project covers application architecture, secure API design, data modelling, state management, and production delivery across the frontend and backend.

Core responsibilities

- Defined functional requirements and translated them into frontend, backend, API, authentication, and data-layer designs.
- Designed architecture across React, Express REST services, authentication middleware, Mongoose models, and MongoDB persistence.
- Engineered JWT-based authentication and authorisation using Bearer tokens, protected routes, bcrypt password hashing, credential verification, and authenticated client-server communication.
- Implemented client-side state management, asynchronous resource orchestration, validation, error handling, and client-server state reconciliation.

- **Software Development Engineer**

API-Driven Application Development

Designed and developed a full-stack system integrating a React frontend, Node.js / Express REST services, MongoDB persistence, and multiple external data providers. Owned implementation across data integration, asynchronous workflows, persistent state, search, external services, and browser-based functionality.

Core responsibilities

- Translated functional requirements into frontend architecture, backend services, API contracts, data models, and integration workflows.
- Designed MongoDB persistence using Mongoose for user libraries, resources, and persistent state.
- Implemented client-side state management, asynchronous resource orchestration, validation, error handling, and client-server state reconciliation.
- Maintained source control, dependency management, linting, testing, troubleshooting, and iterative improvements using Git and GitHub.

- **Full-Stack Software Engineer | AI Engineer**

Knowledge Engineering & Generative AI Application

Designed and engineered a full-stack AI-enabled application integrating a React frontend, Node.js / Express backend, MongoDB persistence, document processing, semantic retrieval, local embedding infrastructure, conversational context, and streamed LLM inference. Held direct ownership of the surrounding application architecture and AI data pipeline rather than limiting the implementation to model API integration

Core responsibilities

- Designed end-to-end architecture spanning frontend, backend services, persistence, document processing, semantic retrieval, and model inference.

- Engineered RAG workflows covering document ingestion, overlapping text chunking, semantic embedding generation, vector persistence, cosine similarity calculation, similarity ranking, and Top-K retrieval.
- Implemented local semantic embedding infrastructure using Hugging Face Transformers, ONNX, and all-MiniLM-L6-v2, including lazy model loading, mean pooling, and embedding generation for knowledge chunks and search queries.
- Developed document processing for TXT, Markdown, PDF, and DOCX content using Multer, PDF Parse, and Mammoth, including multipart uploads, content extraction, validation, normalisation, chunking, and persistence.
- Integrated server-side LLM inference through the Groq SDK and implemented streamed responses using asynchronous provider streams and Server-Sent Events, with incremental client-side parsing, completion handling, error processing, and request cancellation.
- Designed and maintained MongoDB/Mongoose models and REST API workflows for conversations, messages, memories, knowledge documents, embedded vectors, semantic search, document ingestion, and streamed AI inference.

• **Full-Stack Software Engineer | Database Application Engineer**

Enterprise and Database Management

Designed and engineered a full-stack database management system centred on PostgreSQL, SQL, relational modelling, Prisma ORM, and runtime schema management. Owned technical architecture across the browser interface, REST services, database layer, schema management, SQL execution, and metadata synchronisation.

Core responsibilities

- Designed the PostgreSQL architecture and relational models, including primary and foreign keys, unique constraints, nullability, defaults, identity columns, and relationship integrity.
- Developed Express REST services supporting relational CRUD, dynamic record operations, schema management, relationship operations, database health monitoring, and SQL execution.
- Developed a dedicated SQL builder and validation layer for DDL generation, including identifier quoting and escaping, data-type allowlisting, type normalisation, and constraint validation.
- Engineered runtime schema operations supporting table creation and modification of columns, constraints, and relationships through PostgreSQL DDL.
- SQL execution through POST /api/query, with PostgreSQL-aware integrity validation, conflict handling, state verification, and repeatable database seeding.

January 2020 - June 2026

Technical & Digital Operations Lead | Co-Founder Firstdaes Entertainment

Co-founded and operated an events and entertainment business with responsibility spanning digital product and customer experience, technical and digital operations, business systems, workflow design, project delivery, brand strategy, marketing technology, and operational optimisation.

Responsibilities

- Led the development and ongoing optimisation of the company's digital ecosystem, spanning web presence, digital platforms, business tools, analytics, content systems, and technology-enabled workflows.

- Translated business objectives and operational requirements into digital solutions, structured workflows, and practical technology implementations.
- Applied systems thinking to analyse operational processes, identify inefficiencies, and design improved workflows and technology-enabled ways of working.
- Researched, evaluated, configured, and integrated digital tools and third-party services to support business operations, communications, marketing, analytics, and process efficiency
- Managed digital initiatives end-to-end, from requirements definition and solution planning through implementation, testing, troubleshooting, iteration, and optimisation.
- Leveraged data, performance metrics, analytics, and operational feedback to evaluate digital initiatives and drive continuous improvement.
- Worked across web technologies, digital platforms, data and analytics tools, automation, content systems, and integrated services to maintain and evolve the company's digital infrastructure
- Owned technical and operational problem-solving across the company's digital environment, independently researching solutions, resolving issues, and implementing improvements
- Managed multiple concurrent digital and operational projects, coordinating dependencies, external providers, resources, timelines, and delivery requirements

February 2021 - September 2025

Contract Specialist | Technical Content & Documentation **3Play Media**

Worked extensively with technically complex documentation, training material, and technology-focused content across enterprise software, cloud computing, networking, cybersecurity, software development, and emerging technologies. Responsible for transforming and quality-assuring technical material while maintaining terminology accuracy, contextual integrity, structural consistency, and technical clarity across complex subject matter.

Responsibilities

- Reviewed, transcribed, edited, and proofread technical documentation and training content covering software development, cloud infrastructure, networking, cybersecurity, enterprise technology, and emerging technologies.
- Worked extensively with technical terminology, product specifications, software concepts, infrastructure terminology, security concepts, and technology-specific language across diverse subject areas.
- Analysed source material for inconsistencies, ambiguity, terminology conflicts, structural issues, and technical inaccuracies before producing final content.
- Maintained strict quality and consistency standards across technical documentation, including terminology, formatting, structure, instructional clarity, and contextual accuracy.
- Adapted to different technical domains and technology ecosystems, requiring rapid acquisition of domain-specific knowledge and the ability to work effectively with unfamiliar technologies.
- Worked with technical content associated with major enterprise technology organisations and platforms, including Cisco, Oracle, Palo Alto Networks, Unreal Engine, Deloitte, ThousandEyes, Salesforce, NVIDIA, CBTNuggets, and Apple.
- Operated independently across contract-based assignments, managing quality, accuracy, deadlines, and deliverables while working with highly specialised technical material.

Projects

Portfolio: <https://keenosmith.vercel.app>

Multi-Agent AI Systems - Microsoft Foundry / Azure

Agentic AI · Multi-Agent Architecture · OpenAI Models · RAG · Tool Calling · OpenAPI · MCP · Python · Azure RBAC · OpenTelemetry

Multi-agent cloud engineering systems developed with Microsoft Foundry and Azure, using specialised agents, tool-augmented reasoning, RAG, MCP, OpenAPI integrations, structured engineering outputs, infrastructure APIs, asynchronous workflows, policy controls, observability, evaluation and human-in-the-loop execution. The architectures apply agentic AI to software engineering and cloud architecture while maintaining deterministic controls, bounded permissions and auditable decision workflows.

- **AI-Powered Software Engineering & DevOps Operations Platform**

Multi-agent software engineering and DevOps platform separating repository analysis, source-code inspection, application diagnostics and operational investigation into specialised engineering agents. Integrates GitHub repositories, source code, pull requests, documentation, application logs, monitoring telemetry, deployment history and engineering knowledge to analyse software behaviour, investigate failures, correlate evidence and generate structured remediation plans. Agent tool calling, RAG, API integrations and controlled workflow orchestration allow AI agents to operate against real software-development and operational systems rather than functioning as a standalone coding assistant.

GitHub: <https://github.com/keenosmith-del/microsoft-devops-incident-intelligence>

Cloud Security, DevOps & Agentic AI - AWS

Cloud Security · Agentic AI · Event-Driven Architecture · DevOpsSRE · Lambda · DynamoDB · CloudWatch · CodeDeploy · IAM · KMS · Docker · EventBridge · Bedrock · CloudTrail · S3 · SQS · Fargate · EC2

AWS-native cloud engineering platforms combining event-driven architecture, serverless services, containerised delivery, security automation, agentic AI, workflow orchestration, observability, and controlled automated recovery. The projects demonstrate how AWS-native infrastructure and deterministic engineering controls can be combined with Bedrock and AgentCore to build secure, stateful and production-oriented systems across cloud security operations and software delivery reliability.

- **Docker Distributed Developer Deployment Platform**

Docker-first AWS DevOps and SRE platform automating the software delivery lifecycle while continuously evaluating deployed application health and initiating controlled recovery workflows. Uses GitHub, CodePipeline, CodeBuild, Docker, ECR and ECS/Fargate to build, validate, publish and deploy immutable container artefacts, with CloudWatch telemetry feeding EventBridge and Step Functions for deployment-health evaluation and Bedrock/AgentCore-assisted root-cause analysis. Combines deterministic reliability guardrails with bounded AI reasoning to support promotion, hold and rollback decisions, alongside retries, dead-letter queues, idempotent operations, observability and controlled automated recovery.

GitHub: <https://github.com/keenosmith-del/aws-distributed-developer-deployment>

Google Cloud

Cloud-Native Engineering · DevSecOps · Software Supply-Chain Security · Multi-Agent Systems · Gemini · Vertex AI · Firestone · Cloud KMS · BigQuery · Cloud Run · Dataplex · Artifact Registry · Google ADK

Google Cloud-native engineering platforms combining containerised application development, event-driven architecture, specialised AI agents, software supply-chain analysis, enterprise data governance, security controls, and scalable serverless workloads. The projects use Google ADK and Gemini alongside deterministic cloud-native services to build evidence-driven security and governance workflows across software repositories, application dependencies, enterprise data estates, and cloud infrastructure.

- **Autonomous Data Governance & Compliance Platform**

Enterprise data governance control plane designed to continuously discover, classify and assess heterogeneous data assets across Cloud Storage, BigQuery and Firestore. Combines metadata management, Sensitive Data Protection, access analysis, policy evaluation, risk and confidence scoring, and evidence-based AI reasoning to produce structured governance findings and route approved actions through controlled remediation workflows. Uses specialised ADK agents for discovery, classification, policy and governance analysis while maintaining deterministic controls for sensitive-data detection, policy enforcement, IAM evaluation, auditability and human approval.

GitHub: <https://github.com/gcp-data-governance-compliance>

Claude-Powered AI Engineering - Anthropic

LLM Application Engineering · Context Engineering · Semantic Analysis · Anthropic API · JSON · AST Parsing · Knowledge Graphs · AI Evaluation · PostgreSQL · Node.js · RAG · OpenAPI

Claude-powered engineering platforms combining long-context reasoning, structured generation, retrieval, semantic analysis, static code analysis, architecture modelling, and decision workflows. The projects use Claude for evidence-grounded reasoning over complex technical and regulatory information, with deterministic processing, structured data pipelines, evaluation, human review, and locally containerised application infrastructure surrounding the model layer.

- **Claude-Powered Regulatory Change Impact & Control Mapping Platform**

Regulatory analysis platform that processes evolving regulatory documents, identifies material changes between versions, extracts obligations, and maps those changes to organisational controls, policies, and operational processes. Combines document ingestion, semantic comparison, local embeddings, vector retrieval, structured Claude outputs, evidence grounding, impact assessment, control mapping, human review, and remediation planning through a custom React and Python/Node.js application.

GitHub: <https://github.com/keenosmith-del/claude-reg-change-impact>

Workflow Orchestration & Automation - n8n

Event-Driven Architecture · API Integration · Webhooks · TypeScript · RedisDocker · OpenTelemetry · GitHub · Express · Prometheus · Grafana

Workflow orchestration and integration platforms developed with n8n as the event-driven automation layer, combining webhooks, API orchestration, asynchronous processing, state management, reliability patterns, observability, containerised infrastructure, and custom local applications. The projects demonstrate enterprise workflow design, heterogeneous system integration, API operations, fault handling, automated remediation, and production-oriented engineering beyond conventional no-code automation.

- **Webhooks API Orchestrations & Integration Reliability Engine**

API operations platform for continuously validating, monitoring, and operating external integrations through webhook-driven workflows and automated reliability controls. Combines synthetic API transactions, OpenAPI and JSON Schema contract validation, API gateway integration, latency and error analysis, retry and backoff strategies, fault injection, integration state management, automated remediation, and operational observability through a custom React and Node.js console.

GitHub: <https://github.com/keenosmith-del/n8n-webhooks-api-orch>

Full-Stack Software Projects

Full-Stack Engineering · Application Architecture · React · Vite · Node.js · Express · MongoDB · PostgreSQL · Prisma

A portfolio of end-to-end full-stack systems demonstrating frontend architecture, backend service development, REST API design, database architecture, authentication and authorization, state management, asynchronous application flows, component architecture, and integration between client applications, backend services, databases, and external APIs.

Projects span document and relational data models, JWT-based authentication, protected API workflows, Mongoose and Prisma data access, PostgreSQL schema and SQL operations, third-party API integration, external service integration, AI and LLM integration, retrieval workflows, validation, error handling, and application lifecycle management across independently structured frontend and backend systems.

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Certifications and Licenses

September 2026

Google Cloud - Manage Kubernetes in Google Cloud

Intermediate hands-on training in Google Kubernetes Engine (GKE) covering Kubernetes deployments with kubectl, cluster and application management, monitoring and debugging, Prometheus, containerised application deployment, Artifact Registry, Kubernetes manifests, and continuous delivery through practical labs and challenge assessment.

September 2026

n8n - n8n Workflow Automation & AI

Applied n8n workflow engineering across API integrations, HTTP requests, OAuth2 and API authentication, secure webhooks, pagination, data transformation, conditional and parallel execution, sub-workflows, custom JavaScript/Python, workflow state, and error handling. Extended into AI-powered workflows using AI Agents, tools and memory, with practical implementation of testing, debugging, modular workflow design, versioning, monitoring, recovery, and production-readiness patterns.

August 2026

Microsoft - *Microsoft Applied Skills: Microsoft Defender XDR*

Demonstrating practical security operations skills with Microsoft Defender XDR, including incident management, endpoint investigation, threat detection and Advanced Hunting with KQL.

July 2026

Microsoft x Founderz

Microsoft Agent Architect

Cyber Security & Threat Defense

Architect-level training focused on designing scalable AI agent solutions using Microsoft Foundry, including agent architecture, project-based implementation, security labs, simulations, prototyping, and production-oriented approaches to enterprise AI.

May 2026

AWS x Udacity

AWS AI/ML Scholars Programme - AI Practitioner

AWS AI Programmer Nanodegree

Project-based AI/ML development covering Python, NumPy, Pandas, neural networks, PyTorch, transformer architectures, generative AI, prompt engineering, and AWS AI/ML services.

December 2025

WeThinkCode_

Generative AI for Software Engineers

AI Course for Developers

Applied generative AI development covering LLM integration, prompt engineering, AI-assisted software development, agentic workflows, API integration, and incorporating generative AI capabilities into software applications.